

Revised Generation Interconnection System Impact Study Report

For Queue Project Q24-2406

Milvon 100 kV
October 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2406. The Interconnection Customer proposes a 200.0 MW battery energy storage facility interconnecting at the Milvon 100 kV substation in the Illinois area. The request was received on May 06, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,791,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Milvon - Pearson Rd 100.0 kV Cir. 1	Milvon Substation - Pearson Rd 100.0 kV #2	188.8	142	73.0
Pearson Rd - Milvon 100.0 kV Ckt 2	Milvon - Pearson Road 100.0 kV	188.8	142	73.0
Orfordville - Pearson Rd 100.0 kV #2	Orfordville - Pearson Rd 100.0 kV	156.6	145	52.3
Brown Plant - Orfordville Substation 100.0 kV Circuit 1	Orfordville - Brown Plant 100.0 kV L	151.0	189	55.7
Brown Plant - Orfordville 100.0 kV	Orfordville - Brown Plant 100.0 kV #1	149.3	192	55.7
Brown Plant - Lockport 100.0 kV	Brown Plant 230-1 kV Transformer	140.2	120	44.6
Orfordville - Pearson Rd 100.0 kV Ckt 1	Orfordville - Pearson Road 100.0 kV #2	133.7	170	52.3
E Fairfax - Tropic Isle 100.0 kV L	Compo 230/1 kV Transformer	122.8	114	16.2
Kanss - Bollingbrook 100.0 kV line	Brown Plant 100-1 kV Auto	113.1	104	29.1
Pine Ridge - Sarival 100.0 kV line	Compo 230/1 kV Auto	112.9	147	12.2
Bringinghurst - Naguabo 2701 Substation 100.0 kV	Brown Plant 100 kV/1 kV Auto	112.6	109	26.3
Royce - Spring Brook 100.0 kV	Brown Plant 100/1kV Auto	111.7	157	16.1
Oswego Falls W Substation - Route 59 100.0 kV Line 1	Oswego Falls W - Route 59 100.0 kV #2	111.4	164	5.9
Brown Plant 100/1kV Transformer Bank	Lincoln 100/1 kV TX	109.7	273	28.3
Milvon - Kalispell 100.0 kV	Compo 100-1 kV Auto	109.1	148	21.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wyldwood - Bogard 100.0 kV Cir. 2	Bogard - Wyldwood Substation 100.0 kV	108.8	146	7.8
Lockport - Buckeye Canada 100.0 kV line	Brown Plant 230/1 kV XFMR	108.4	108	24.3
Ft StVrain - Goudchaux 100.0 kV #1	Goudchaux - Ft St Vrain 100.0 kV Cir. 2	106.9	157	5.7
Locust - Goudchaux 100.0 kV Ckt 1	Goudchaux - Locust 100.0 kV Cir. 2	105.8	154	6.0
Goudchaux - Locust 100.0 kV Circuit 2	Locust - Goudchaux 100.0 kV Cir. 1	105.8	154	6.0
Goudchaux - Locust 100.0 kV #3	Locust - Goudchaux 100.0 kV Circuit 2	105.8	154	6.0
Wyldood - Otay Mesa 100.0 kV Line 1	Wyldwood - Otay Mesa 100.0 kV Ckt 3	105.5	157	7.9
Goudchaux - Fort St Vrain 100.0 kV Line	Goudchaux - Ft St Vrain 100.0 kV Line 1	104.4	161	5.7
W Kingston - Shell Chemical 100.0 kV #2	W Kingston - Shell Chemical 100.0 kV	104.4	154	8.4
Goudchaux Substation - Fort St Vrain 100.0 kV	Goudchaux - Ft St Vrain 100.0 kV Ckt 1	104.2	161	5.7
Mason St - New Departure 230.0 kV	Beaver - Trego 500.0 kV	103.3	1,652	7.6
Bogard - Wyldwood 100.0 kV Line 1	Bogard - Wyldwood 100.0 kV Cir. 2	103.3	154	7.8
Pickford - Herald-Twin Cities 230.0 kV	Trego - Beaver 500.0 kV	102.1	2,114	9.4
Post Rock - Myrtle Point Substation 230.0 kV	Trego - Beaver 500.0 kV	101.1	2,995	12.4
Post Rock - Newport Tie 230.0 kV	Trego - Beaver 500.0 kV	101.1	2,963	12.8
Canal Rd - Weedpatch 230.0 kV	Trego - Beaver 500.0 kV L	100.1	1,782	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Milvon - Pearson Rd 100.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,572,000	100% (\$1,572,000)
Pearson Rd - Milvon 100.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,644,000	100% (\$1,644,000)
Orfordville - Pearson Rd 100.0 kV #2	Rebuild line with higher-capacity conductor	\$1,442,000	87% (\$1,251,000)
Orfordville - Brown Plant 100.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$447,000	87% (\$388,000)
Brown Plant - Orfordville 100.0 kV	Rebuild line with higher-capacity conductor	\$591,000	87% (\$513,000)
Brown Plant - Lockport 100.0 kV	Rebuild line with higher-capacity conductor	\$2,193,000	87% (\$1,906,000)
Orfordville Substation - Peason Road 100.0 kV L	Rebuild line with higher-capacity conductor	\$2,430,000	87% (\$2,110,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Fairfax - Tropic Isle 100.0 kV	Rebuild line with higher-capacity conductor	\$879,000	83% (\$728,000)
Kanss - Bollingbrook 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,593,000	90% (\$3,242,000)
Sarival - Pine Rige 100.0 kV	Rebuild line with higher-capacity conductor	\$1,280,000	61% (\$782,000)
Bringhurst - Naguabo 2701 Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$3,967,000	73% (\$2,886,000)
Royce - Spring Brook 100.0 kV	Rebuild line with higher-capacity conductor	\$1,715,000	100% (\$1,715,000)
Route 59 - Oswego Falls W 100.0 kV	Rebuild line with higher-capacity conductor	\$1,557,000	100% (\$1,557,000)
Brown Plant 100 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$1,969,000	75% (\$1,476,000)
Milvon - Kalispell 100.0 kV	Reconductor line	\$1,372,000	100% (\$1,372,000)
Bogard - Wyldwood 100.0 kV Cir. 2	Reconductor line	\$781,000	100% (\$781,000)
Lockport - Buckeye Canada 100.0 kV line	Reconductor line	\$726,000	87% (\$631,000)
Goudchaux - Ft St Vrain 100.0 kV	Reconductor line	\$1,703,000	100% (\$1,703,000)
Lcust - Goudchaux Substation 100.0 kV Cir. 1	Reconductor line	\$714,000	100% (\$714,000)
Locust - Goudchaux 100.0 kV Cir. 2	Reconductor line	\$377,000	100% (\$377,000)
Locust - Goudchaux 100.0 kV Circuit 3	Reconductor line	\$656,000	100% (\$656,000)
Wyldwood - Otay Mesa 100.0 kV Cir. 1	Reconductor line	\$899,000	100% (\$899,000)
Goudchaux - Fort St Vrain 100.0 kV Line	Reconductor line	\$2,348,000	100% (\$2,348,000)
W Kingston - Shell Chemical 100.0 kV Cir. 2	Reconductor line	\$1,253,000	67% (\$834,000)
Goudchaux Substation - Fort St Vrain 100.0 kV	Reconductor line	\$1,882,000	100% (\$1,882,000)
New Departure - Mason St 230.0 kV	Reconductor line	\$15,354,000	2% (\$294,000)
Wldwood - Bogard 100.0 kV line	Reconductor line	\$496,000	100% (\$496,000)
Pickford - Herald-Twin Cities 230.0 kV	Reconductor line	\$29,523,000	2% (\$564,000)
Myrtle Point - Post Rock Substation 230.0 kV	Reconductor line	\$9,358,000	2% (\$178,000)
Post Rock - Newport Tie Substation 230.0 kV	Reconductor line	\$9,254,000	2% (\$176,000)
Canal Rd - Weedpatch 230.0 kV	Reconductor line	\$6,063,000	2% (\$116,000)
	Total allocated to project		\$35,791,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.